

Assess learning and plan transition

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: A cuboid has volume 756 cm^3 , length 12 cm and width 7 cm. Find its height. Copy or complete the first step.

2. Calculate $2 \frac{1}{4}$ divided by 3, then write the answer as a decimal.

3. Solve $5x + 8 = 73$ and convert 3.45 km to metres.

4. Miro says: Miro records a general target such as 'get better at maths'. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: A cuboid has volume 756 cm^3 , length 12 cm and width 7 cm. Find its height. Copy or complete the first step.
Answer 9 cm.
2. Calculate $2 \frac{1}{4}$ divided by 3, then write the answer as a decimal.
Answer $\frac{3}{4}$ or 0.75.
3. Solve $5x + 8 = 73$ and convert 3.45 km to metres.
Answer $x = 13$ and 3,450 m.
4. Miro says: Miro records a general target such as 'get better at maths'. Is this correct? Explain.
Answer Use assessment evidence to name the exact concept, method, example and support required next.

Assess learning and plan transition

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. A cuboid has volume 756 cm^3 , length 12 cm and width 7 cm . Find its height.

6. Check this result using an inverse, estimate or known fact: Solve $5x + 8 = 73$ and convert 3.45 km to metres.

2. Calculate $2 \frac{1}{4}$ divided by 3, then write the answer as a decimal.

7. Prove or explain your reasoning: Find the area of a triangle with base 18 cm and height 11 cm , then find the mean of 18, 22, 25 and 15.

3. Solve $5x + 8 = 73$ and convert 3.45 km to metres.

8. Identify and correct this misconception: Miro records a general target such as 'get better at maths'.

4. Solve this independently and show every stage: A cuboid has volume 756 cm^3 , length 12 cm and width 7 cm . Find its height.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Calculate $2 \frac{1}{4}$ divided by 3, then write the answer as a decimal.

10. Write and solve a similar question to: Solve $5x + 8 = 73$ and convert 3.45 km to metres.

Teacher answers and checking prompts

1. A cuboid has volume 756 cm³, length 12 cm and width 7 cm. Find its height.
Answer 9 cm.
2. Calculate $2\frac{1}{4}$ divided by 3, then write the answer as a decimal.
Answer $\frac{3}{4}$ or 0.75.
3. Solve $5x + 8 = 73$ and convert 3.45 km to metres.
Answer $x = 13$ and 3,450 m.
4. Solve this independently and show every stage: A cuboid has volume 756 cm³, length 12 cm and width 7 cm. Find its height.
Answer 9 cm.
5. Use a second method or representation for: Calculate $2\frac{1}{4}$ divided by 3, then write the answer as a decimal.
Answer Expected result: $\frac{3}{4}$ or 0.75. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Solve $5x + 8 = 73$ and convert 3.45 km to metres.
Answer Expected result: $x = 13$ and 3,450 m. A valid check must be shown.
7. Prove or explain your reasoning: Find the area of a triangle with base 18 cm and height 11 cm, then find the mean of 18, 22, 25 and 15.
Answer 99 cm² and 20. The explanation must show why the method works.
8. Identify and correct this misconception: Miro records a general target such as 'get better at maths'.
Answer Use assessment evidence to name the exact concept, method, example and support required next.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Solve $5x + 8 = 73$ and convert 3.45 km to metres.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Assess learning and plan transition

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Prove or explain your reasoning: Find the area of a triangle with base 18 cm and height 11 cm, then find the mean of 18, 22, 25 and 15.

6. Create a new example that tests the same idea as: Solve $5x + 8 = 73$ and convert 3.45 km to metres.

2. Prove the answer to this question, rather than only calculating it: A cuboid has volume 756 cm³, length 12 cm and width 7 cm. Find its height.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Calculate $2\frac{1}{4}$ divided by 3, then write the answer as a decimal.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: $x = 13$ and 3,450 m.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro records a general target such as 'get better at maths'.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: Find the area of a triangle with base 18 cm and height 11 cm, then find the mean of 18, 22, 25 and 15.
Answer 99 cm² and 20. The explanation must show why the method works.
2. Prove the answer to this question, rather than only calculating it: A cuboid has volume 756 cm³, length 12 cm and width 7 cm. Find its height.
Answer Expected result: 9 cm. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Calculate $2\frac{1}{4}$ divided by 3, then write the answer as a decimal.
Answer Expected result: $\frac{3}{4}$ or 0.75. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: $x = 13$ and 3,450 m.
Answer One valid reconstruction is: Solve $5x + 8 = 73$ and convert 3.45 km to metres. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro records a general target such as 'get better at maths'.
Answer Use assessment evidence to name the exact concept, method, example and support required next.
6. Create a new example that tests the same idea as: Solve $5x + 8 = 73$ and convert 3.45 km to metres.
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.